2018 Ph H2 Q15

Section: Electricity

Topic: Current, PD, Power, Resistance

A resistor is connected to a power supply.
The potential difference (p.d.) across the resistor is measured as the current is varied.
The following results are obtained:

Current (A)	Voltage (V)
0.20	1.00
0.40	2.00
0.60	3.00
0.80	4.00
1.00	5.00

(a) Plot a graph of voltage (y-axis) against current (x-axis).

✅ The data is perfectly linear:
Plot a straight line passing through the origin (0,0) up to (1.00, 5.00).

(b) Determine the resistance of the resistor.

Since the graph is a straight line through the origin:

$$\text{Resistance} = \frac{V}{I} = \frac{5.00}{1.00} = 5.0\ \Omega$$

Or:
The gradient of the line = resistance.

(c) The resistor is now replaced with one that behaves differently.

The graph of voltage against current is now **curved**.
What does this indicate about the resistance of the new resistor?

✅ A curved graph means the **resistance is not constant**.
Specifically:

- If the curve becomes steeper, resistance is **increasing** with current (e.g. a filament lamp).
- This is **non-ohmic** behaviour — the resistor does **not obey Ohm's Law**.

🧠 Revision Tips

- **Ohm's Law**: $V = IR$ — applies to **ohmic conductors** (straight-line graph)
- Slope of V vs I graph = **resistance**
- A **curved** graph = **non-ohmic** behaviour, e.g. filament bulb, thermistor
- Resistance varies with **temperature** in many components
